The conversion among the three different forms is heavily influenced by pH. In acidic solution the amino acid behaves like a protonated derivative and moves towards cathode under an electric field. In an alkaline medium the same amino acid behaves like an anion and moves towards anode.

At a particular pH, the net charge of an amino acid becomes neutral (Form II) and such ionic forms are known as zwitter ions and the corresponding pH is known as its iso electric point (pI). The titration curve of alanine is given in figure 3.5 Note that since there are two ionisable groups in alanine, it has two pKa values denoted as pKa_1 which is equal to 2.3 and pKa_2 which is equal to 9.1. The iso electric point is denoted as pI and for alanine it is 5.7.

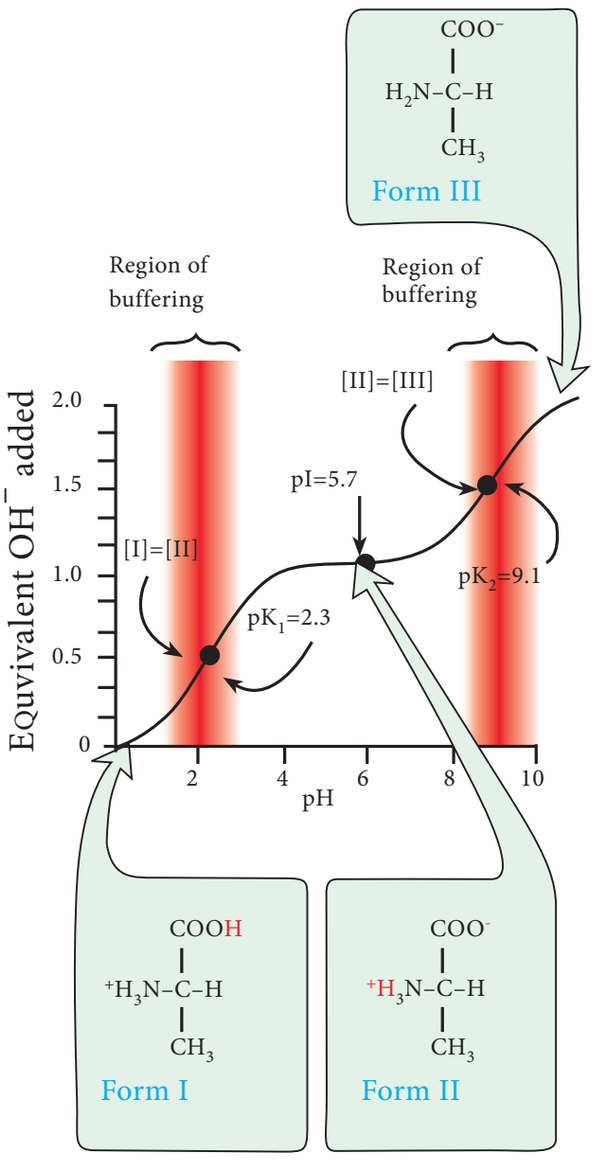

Figure 3.5: The titration curve of alanine

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